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UNIVERSITÉ D'OTTAWA · UNIVERSITY OF OTTAWA

Week 3 — Training Day 2: Mission Operations Training

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Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Classification: Internal

Training Day 2 — Mission Operations Training

Day at a glance

When	What	Output
09:00 – 10:00	Voice-loop discipline + GO/NO-GO protocol	Voice card learnt
10:00 – 11:30	Procedure walk-through: NOM-001 and CTG-003	Q&A captured
11:30 – 12:00	Shift handover practice (NOM-012)	Two handovers logged
13:00 – 14:00	Flight-dynamics workshop	Pass timing predicted
14:00 – 15:30	Dry-run 1 — nominal pass, no failures	Operator-shift logs
15:30 – 16:30	Dry-run 2 — nominal pass with one failure injected	Anomaly tickets, debrief
16:30 – 17:00	**Operations training quiz**	Quiz submitted

Setting: uOttawa Cyberrange. Two consoles set up by 09:00. Both dry-runs use the **Nominal Post-Commissioning** breakpoint (BP-1).

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1. Voice-loop discipline

[Claude Code: 350 words. Cover the canonical phrasing used at NASA and ESA mission control. Include a one-page reference card with: addressing pattern ("FD, AOCS — go for star-tracker enable"), GO/NO-GO call-and-response, hot-mike rules, "loop discipline" expectations, the rule that the FD does not command directly. Cite ESA *Operations Manual for Mission Control* and the Mission Operations Voice Procedure standard from NASA APPEL — https://appel.nasa.gov/

Also provide a short EN/FR glossary of voice-loop callouts. uOttawa is bilingual, but the callouts themselves stay in English for every crew; the glossary explains their meaning in French.]

1.1 Standard callouts

[Claude Code: produce a 12-row table — callout · meaning · response. Examples: "Comm check 1-2-3", "Standby plus 30", "TTC report", "AOS in 2", "Pass closed".]

1.2 GO/NO-GO protocol

[Claude Code: 150 words. The FD polls each position; each position returns GO, NO-GO with reason, or HOLD with reason. The FD makes the decision. No commanding without an explicit FD authorisation.]

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2. Procedure walk-through

2.1 NOM-001 Pass Startup

[Claude Code: walk students through procedure `procedures/nominal/startup.md` step by step. Show how each step maps to MCS UI actions. ~350 words plus annotated MCS screenshots.]

2.2 CTG-003 TTC Link Loss Recovery

[Claude Code: walk students through `procedures/contingency/ttc_link_loss.md`. Emphasise the detection/assessment/recovery loop. ~350 words plus annotated MCS screenshots showing the FDIR panel and link-status telemetry.]

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3. Shift handover practice (NOM-012)

[Claude Code: 150 words. Use procedure `nominal/shift_handover.md`. Each team performs two handovers — outgoing FD briefs incoming FD using the NOM-012 template — and the instructor times them. Target: under 6 minutes.]

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4. Flight-dynamics workshop

[Claude Code: 400 words plus three Python figures. Workshop exercises:

1. Read the orbit (TLE in `configs/eosat1/orbit.yaml`); compute period and rev/day; predict the next 24 h of Iqaluit AOS times.
2. Compute pass duration as a function of maximum elevation; show the typical 8–12 minute polar pass.
3. Explain eclipse: where it starts in the orbit, how long it lasts, what the power and thermal consequences are.

Figures (use the brand stylesheet from `outputs/sample_figures/`):

- Ground-track + Iqaluit and Troll visibility cones
- Pass-elevation profile vs time for a typical Iqaluit pass
- Eclipse fraction over a 24 h window]

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5. Dry-runs

Dry-run 1 — nominal pass

- **Scenario:** BP-1 (Nominal Post-Commissioning).
- **Duration:** 90 min real time, 90 min sim time (1× speed).
- **Procedures expected:** NOM-001 → NOM-002 (imaging) → NOM-003 (data downlink) → NOM-009 (routine health check) → LOS.
- **Failure injection:** none.
- **Outputs:** operator shift log; observer team uses *Communications* peer-review worksheet only.

Dry-run 2 — nominal pass with one failure

- **Scenario:** BP-1 with one easy failure injected ~T+30 min.
- **Recommended injection:** `eps_overcurrent.yaml` on the `payload` line (forces CTG-012 *Overcurrent Response*).
- **Operator pass criterion:** anomaly detected, isolated, and procedure CTG-012 executed; pass continues to LOS.
- **Outputs:** operator shift log + anomaly ticket; observer team uses *Technical* peer-review worksheet.

6. Operations training quiz

[Claude Code: produce a 15-question MCQ + short-answer paper, 30 minutes. Topics:

- Voice-loop discipline (3 Qs)
- MCS UI navigation (3 Qs)
- Telemetry interpretation — limits and SIDs (3 Qs)
- Procedure structure — what's in a procedure file (2 Qs)
- Anomaly response steps (2 Qs)
- Pass timing — AOS / TCA / LOS arithmetic (2 Qs)

Solutions on a separate page for the instructor pack.]

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7. Expected reading for Wednesday

> **Expected reading before Wednesday LEOP day.** Read sections 1 – 4 of the *LEOP Day Instructor Pack* student-facing extract (a redacted version of `sim_leop_instructor.md`). Approximate reading time 30 minutes. Read procedures **LEOP-001, LEOP-002, LEOP-006, LEOP-007** in full (`procedures/leop/*.md`). Approximate reading time 30 minutes.

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8. Tools, files & references

Simulator files referenced

- `configs/eosat1/procedures/nominal/startup.md`
- `configs/eosat1/procedures/contingency/ttc_link_loss.md`
- `configs/eosat1/procedures/nominal/shift_handover.md`
- `configs/eosat1/procedures/leop/*.md` (for evening reading)
- `configs/eosat1/scenarios/eps_overcurrent.yaml`

Open references

- NASA APPEL — Voice loop and mission ops standards — https://appel.nasa.gov/
- ESA Operations *Mission Operations Concept* — https://www.esa.int/Enabling_Support/Operations
- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- *NASA Systems Engineering Handbook* (SP-2016-6105 Rev 2) §8 — https://www.nasa.gov/reference/systems-engineering-handbook/
- CCSDS space packet standards (background; PUS itself is ECSS-E-ST-70-41C) — https://public.ccsds.org/Publications/AllPubs.aspx